

Chapter-1

Information Security

1. Introduction to Information Security
2. Types of Information Security
3. Development of Information Security
4. Need for Information Security
5. Threats to Information Security
6. Cyber security and security risk analysis

1. INTRODUCTION TO INFORMATION SYSTEM (IS)

An **Information System (IS)** is an organized system that collects, processes, stores, and distributes information to support **decision-making, coordination, and control** in an organization.

Real-Life Example

- Banking system → processes transactions
- College ERP → manages attendance, marks

Components of IS

1. Hardware

- Physical devices (CPU, keyboard, servers)

2. Software

- System software (OS) + Application software

3. Data

- Raw facts → processed into meaningful information

4. People

- Users, IT professionals, analysts

5. Procedures

- Rules for operating the system

Characteristics of Good IS

- Accurate
- Timely
- Relevant
- Secure

- Efficient

Functions of IS

1. Input (collect data)
2. Processing (analyze)
3. Output (generate reports)
4. Storage (save data)
5. Feedback (improve system)

2. TYPES OF INFORMATION SYSTEMS

1. Transaction Processing System (TPS)

- Handles **day-to-day operations**
- High speed & accuracy required
- Example: Railway reservation

2. Management Information System (MIS)

- Converts data into reports
- Helps middle management

3. Decision Support System (DSS)

- Uses **data analysis & models**
- Helps in complex decisions

4. Executive Information System (EIS)

- Used by top executives
- Provides summarized information

◆ 5. Office Automation System (OAS)

- Improves office productivity

Comparison Table

System	Users	Purpose
TPS	Operational	Daily tasks
MIS	Middle Mgmt	Reports
DSS	Analysts	Decision making
EIS	Top Mgmt	Strategic decisions

3. DEVELOPMENT OF INFORMATION SYSTEMS

System Development Life Cycle (SDLC)

1. Planning Phase

- Define problem
- Feasibility study:
 - Technical
 - Economic
 - Operational

2. Analysis Phase

- Gather requirements
- Tools:
 - Interviews
 - Questionnaires

3. Design Phase

- System architecture
- Database design
- UI design

4. Implementation Phase

- Coding
- Testing:
 - Unit testing
 - Integration testing

5. Maintenance Phase

- Bug fixing
- Updates

SDLC Models

1. Waterfall Model

- Linear approach
- Simple but inflexible

2. Agile Model

- Iterative & flexible
- Frequent updates

3. Spiral Model

- Risk-focused
- Used for large projects

4. INTRODUCTION TO INFORMATION SECURITY

Information Security protects data from:

- Unauthorized access
- Disclosure
- Modification
- Destruction

CIA TRIAD

Confidentiality

- Only authorized users can access data

Integrity

- Data should not be altered

Availability

- Data must be accessible when needed

Additional Principles

- Authentication
- Authorization
- Accountability

5. NEED FOR INFORMATION SECURITY

Reasons

1. Data Protection

- Prevent data theft

2. Financial Security

- Avoid fraud

3. Privacy Protection

- Personal data safety

4. Legal Compliance

- Laws like IT Act

5. Business Continuity

- Avoid downtime

6. THREATS TO INFORMATION SYSTEMS

A threat is any potential danger that can harm a system.

Types of Threats

1. Natural Threats

- Earthquake
- Flood

2. Human Threats

a. Intentional

- Hackers
- Cyber criminals

b. Unintentional

- Human errors

3. Technical Threats

- Software bugs
- Hardware failure

Major Cyber Threats

Malware

- Virus → attaches to files
- Worm → spreads automatically
- Trojan → disguised program

Phishing

- Fake emails to steal data

Ransomware

- Locks data → asks for money

DoS Attack

- Overloads server

MITM Attack

- Attacker intercepts communication

7. INFORMATION ASSURANCE (IA)

Ensures that information is:

- Reliable
- Secure
- Available

Core Elements

1. Authentication

- Verify identity

2. Authorization

- Grant access rights

3. Integrity

- Data correctness

4. Non-repudiation

- Cannot deny actions

Difference: Security vs Assurance

Security	Assurance
Protection	Confidence
Prevents attacks	Ensures reliability

8. CYBER SECURITY

Protection of digital systems from cyber attacks.

Domains

1. Network Security

- Protect networks

2. Application Security

- Secure software

3. Cloud Security

- Protect cloud data

4. Endpoint Security

- Secure devices

Security Tools

◆ Firewall

- Blocks unauthorized access

◆ Antivirus

- Detects malware

◆ Encryption

- Converts data into unreadable form

◆ IDS/IPS

- Detects intrusions

Cyber Security Challenges

- Increasing cyber attacks
- Lack of awareness
- Rapid technology growth

9. SECURITY RISK ANALYSIS

Process of identifying and reducing risks.

Key Terms

Asset

- Valuable resource (data, system)

Threat

- Potential danger

Vulnerability

- Weakness in system

Steps in Risk Analysis

1. Identify Assets

- Data, hardware

2. Identify Threats

- Malware, hackers

3. Identify Vulnerabilities

- Weak passwords

4. Risk Assessment

- High / Medium / Low

5. Risk Control

Risk Mitigation Methods

- Avoid risk
- Reduce risk
- Transfer risk (insurance)
- Accept risk

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